
Learning Long-Horizon Whole-body Manipulation on Unitree G1 with GR00T N1.7

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Abstract

This project investigates long-horizon language-conditioned whole-body manipulation on the Unitree G1 humanoid robot using the NVIDIA GR00T N1.7 Vision-Language-Action (VLA) foundation model. We consider a representative manipulation task in which the robot is required to pick up a toy doll from a tabletop and place it onto a chair according to a natural language instruction. To adapt the pretrained policy to the target task, we collect 128 high-quality whole-body teleoperation demonstrations using the PICO VR teleoperation system and convert them into the LeRobot dataset format for supervised post-training. The fine-tuned policy is deployed on the real robot through the SONIC whole-body controller, which translates the predicted latent actions into executable motions while maintaining whole-body stability. Real-world experiments demonstrate an 80% pickup success rate and a 70% end-to-end task success rate under randomized initial object configurations. Our results show that task-specific fine-tuning effectively transfers a general-purpose humanoid foundation model to a real-world manipulation task, while also revealing remaining challenges including sensitivity to initial conditions, the lack of explicit grasp-state awareness, and long-horizon error accumulation.

1 Introduction

Humanoid robots have emerged as an important platform for embodied artificial intelligence due to their ability to perform manipulation tasks in human-centric environments. Compared with fixed-base manipulators, humanoid robots possess substantially more degrees of freedom and must coordinate perception, locomotion, manipulation, and whole-body balance simultaneously, making policy learning and deployment significantly more challenging.

Recent Vision-Language-Action (VLA) foundation models have shown promising generalization by jointly modeling visual observations, language instructions, and robot actions within a unified policy. Instead of training task-specific controllers from scratch, these models are pretrained on large-scale robotic datasets and can be adapted to downstream tasks through supervised fine-tuning. Among them, NVIDIA GR00T N1.7 provides an open foundation model designed for humanoid manipulation.

Despite this progress, transferring VLA models from offline datasets to real humanoid robots remains challenging. Long-horizon manipulation requires robust perception, stable whole-body control, and reliable execution across multiple sequential stages, where small errors can accumulate and eventually lead to task failure.

In this project, we investigate a representative long-horizon manipulation task on the Unitree G1 humanoid robot. Given ego-view observations, robot proprioception, and a language instruction, the robot is required to pick up a toy doll from a table and place it onto a nearby chair. We collect expert demonstrations using the PICO VR whole-body teleoperation system [NVIDIA and NVLabs \[2026\]](#), convert the data into the LeRobot format, and fine-tune GR00T N1.7 for task-specific adaptation. During deployment, the predicted latent actions are executed through the SONIC whole-body controller on the physical robot.

Experimental results demonstrate that the fine-tuned policy successfully performs the complete manipulation pipeline in real-world environments while also revealing remaining challenges such as sensitivity to initial conditions, limited grasp-state awareness, and long-horizon error accumulation.

The main contributions of this project are summarized as follows:

- We develop a complete real-world manipulation pipeline by integrating GR00T N1.7 with the SONIC whole-body controller on the Unitree G1 humanoid robot.
- We collect 128 high-quality whole-body teleoperation demonstrations, convert them into the LeRobot format, and fine-tune GR00T N1.7 for a long-horizon language-conditioned manipulation task.
- We validate the proposed system through comprehensive real-world experiments and analyze its effectiveness, failure modes, and remaining limitations.

2 Related Work

2.1 Humanoid Manipulation

Humanoid robots have become an important research platform for embodied artificial intelligence because they are capable of performing manipulation tasks in human-centric environments while simultaneously coordinating locomotion, balance, and dexterous manipulation. Compared with fixed-base robotic manipulators, humanoid robots possess substantially more degrees of freedom and must satisfy additional constraints on body stability, collision avoidance, and motion feasibility. Consequently, language-conditioned whole-body manipulation has emerged as a representative benchmark for evaluating the integration of perception, reasoning, and robot control [Tong et al. \[2024\]](#).

Recent progress in robot learning has significantly improved the manipulation capability of humanoid platforms through large-scale demonstration learning and foundation models. Nevertheless, deploying these policies on real robots remains challenging due to the complexity of whole-body coordination, the long-horizon nature of manipulation tasks, and the inevitable distribution gap between offline datasets and real-world environments.

2.2 Vision-Language-Action Foundation Models

Vision-Language-Action (VLA) models formulate robotic manipulation as a multimodal sequence modeling problem, where visual observations, language instructions, and robot states are jointly encoded to predict future robot actions [Zitkovich et al. \[2023\]](#), [Kim et al. \[2024\]](#), [Octo Model Team et al. \[2024\]](#). Benefiting from large-scale robot datasets and pretrained vision-language models, VLA policies have demonstrated impressive generalization across manipulation tasks while significantly reducing the amount of task-specific engineering required by conventional robotic pipelines.

Among recent VLA systems, NVIDIA GR00T introduces an open foundation model designed specifically for humanoid robots [NVIDIA \[2026\]](#). GR00T N1.7 integrates visual observations, language instructions, and robot proprioception to generate latent action representations that can be adapted to downstream tasks through supervised post-training. Owing to its open-source training pipeline and support for the LeRobot dataset format, GR00T provides an attractive framework for investigating task-specific adaptation on real humanoid robots.

2.3 Whole-body Control and Real-world Deployment

Although VLA models can generate high-level manipulation actions, these predictions cannot be directly executed on humanoid robots. Practical deployment requires an additional whole-body controller capable of converting the predicted action representations into dynamically feasible robot motions while satisfying constraints on balance, joint limits, and motion smoothness.

To bridge this gap, NVIDIA introduces SONIC, a latent whole-body control framework integrated into the GR00T deployment pipeline [NVIDIA GEAR and NVLabs \[2026\]](#). Instead of directly predicting low-level joint trajectories, GR00T outputs latent motion representations that are decoded by SONIC into executable whole-body movements. This hierarchical design separates high-level task

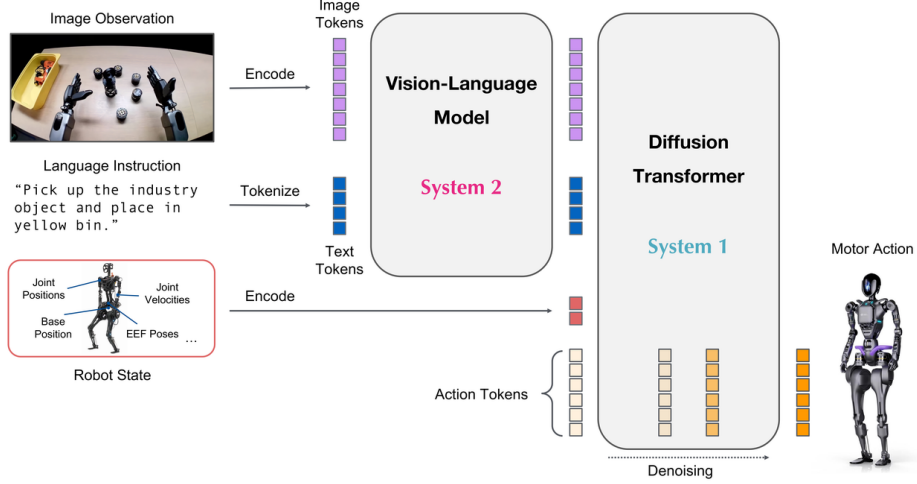


Figure 1: Overview of the GR00T N1.7 framework. Visual observations, language instructions, and robot states are jointly processed to predict latent action representations for language-conditioned humanoid manipulation.

reasoning from low-level motion generation, enabling more stable deployment on complex humanoid platforms.

Following this paradigm, our system adopts GR00T N1.7 as the high-level Vision-Language-Action policy and employs SONIC as the downstream whole-body controller for execution on the Unitree G1 robot. This architecture allows us to focus on task-specific adaptation through demonstration fine-tuning while leveraging a robust motion generation framework for real-world deployment.

3 Method

3.1 Overall Framework

Given an ego-view image, a language instruction, and the current robot proprioceptive state, the GR00T N1.7 policy predicts a sequence of latent actions describing the desired future whole-body motions. These latent actions are subsequently decoded by the SONIC controller into executable robot commands, which are finally executed on the Unitree G1 humanoid robot.

Unlike conventional end-to-end joint-space policies, our system follows a hierarchical architecture that separates high-level task reasoning from low-level motion generation. GR00T is responsible for multimodal perception and action prediction, while SONIC guarantees that the generated motions satisfy the dynamic constraints required for stable whole-body execution. This design enables efficient adaptation through supervised fine-tuning while preserving the robustness of the pretrained whole-body controller.

3.2 Demonstration Collection

Since no public dataset is available for the target manipulation task, we collect expert demonstrations using the PICO VR Whole-body Teleoperation system provided by the GR00T-WholeBodyControl framework. During teleoperation, an operator controls the Unitree G1 robot through a VR headset and handheld controllers, while the framework continuously maps the estimated human body pose to feasible robot motions.

Each demonstration corresponds to a complete manipulation episode consisting of four sequential stages:

1. approaching the target object
2. grasping and lifting the toy doll

3. transporting the object toward the chair
4. placing the object onto the chair

To improve the quality of the collected dataset, every trajectory is manually inspected after recording. Demonstrations containing unsuccessful grasps, interrupted executions, severe visual occlusions, corrupted recordings, or unsafe robot motions are discarded. Only complete demonstrations that successfully accomplish the entire manipulation task are retained for subsequent training.

3.3 Dataset Processing

The collected demonstrations are converted into the LeRobot dataset format required by the official GR00T post-training pipeline. Each episode consists of synchronized multimodal observations, including ego-view RGB images, robot proprioceptive states, language instructions, and action labels.

The dataset is organized into three components:

- **Meta information:** including task definitions and dataset metadata
- **Video recordings:** storing synchronized ego-view observations
- **Robot trajectories:** containing aligned robot states and action labels

This standardized representation enables direct compatibility with the official GR00T fine-tuning framework without additional preprocessing.

3.4 Task-specific Fine-tuning

The processed dataset is used to fine-tune the pretrained GR00T N1.7 foundation model through supervised learning. During training, the model receives three input modalities:

- ego-view RGB observations
- natural language task instructions
- robot proprioceptive states

The output of the policy is a sequence of latent action chunks representing future whole-body motions. Each predicted action consists of a 64-dimensional SONIC latent motion representation together with two 7-dimensional hand joint targets for controlling the left and right grippers, resulting in a total action dimension of 78.

Following the official GR00T post-training configuration, the language model backbone is frozen during fine-tuning, while the remaining trainable modules are optimized using supervised behavior cloning. Training is performed on eight NVIDIA H200 GPUs for 60K optimization steps.

3.5 Real-world Deployment

During deployment, the fine-tuned policy continuously predicts latent action chunks conditioned on the latest visual observations, language instruction, and robot state. Instead of directly commanding robot joints, the predicted latent representations are passed to the SONIC controller, which generates dynamically feasible whole-body motions while enforcing constraints on joint limits, balance, and motion smoothness.

This hierarchical execution strategy allows the high-level policy to focus on language-conditioned manipulation while delegating low-level whole-body control to SONIC. Consequently, the resulting system can execute long-horizon manipulation tasks on the Unitree G1 robot with stable whole-body coordination in real-world environments.

4 Experiments

4.1 Experimental Setup

We evaluate the proposed system through both offline prediction and real-world deployment. Offline evaluation measures the prediction accuracy of the learned latent actions on held-out demonstrations,

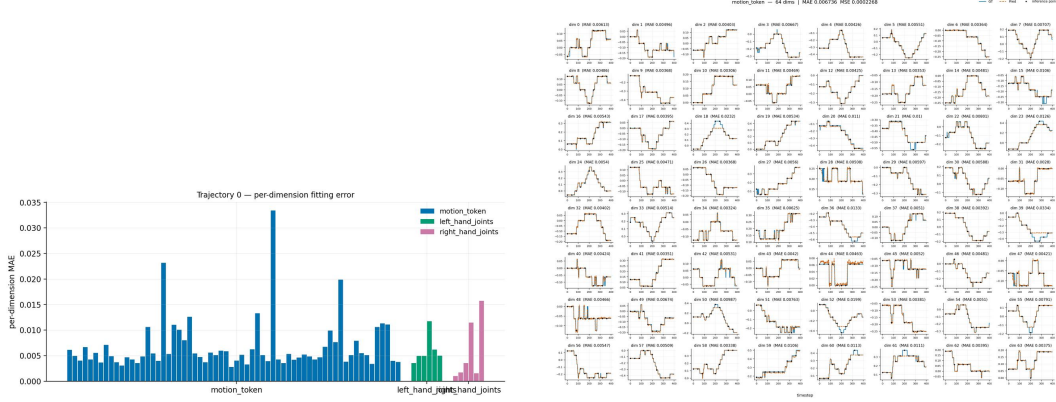


Figure 2: Offline evaluation on the validation set. Left: per-dimension prediction errors. Right: representative comparisons between predicted and ground-truth latent actions.

while real-world evaluation assesses the robot’s ability to complete the manipulation task on the physical Unitree G1 platform.

The policy is fine-tuned following the official GR00T post-training pipeline. During training, the LLM backbone is frozen and the remaining modules are optimized using supervised behavior cloning. Offline performance is evaluated using Mean Squared Error (MSE) and Mean Absolute Error (MAE), while real-world evaluation reports both the pickup success rate and the end-to-end task success rate.

Table 1: Fine-tuning configuration.

Item	Value
Base model	GR00T N1.7
Training objective	Supervised fine-tuning
Frozen module	LLM backbone
GPUs	8× NVIDIA H200
Training steps	60K
Global batch size	1024
Prediction horizon	40
Action dimension	78
Training time	~14 h

4.2 Offline Evaluation

We first evaluate the fine-tuned policy on held-out demonstrations using open-loop prediction metrics. Table 2 reports the prediction accuracy for different action components. The model achieves consistently low prediction errors across both the SONIC latent actions and the hand joint targets, indicating successful imitation of the demonstrated manipulation behaviors.

Table 2: Open-loop prediction performance on the validation set.

Component	Dim.	MSE	MAE
SONIC latent motion	64	0.0002268	0.0067361
Left hand	7	0.0018883	0.0051907
Right hand	7	0.0002471	0.0049951
Overall	78	0.0003777	0.0064411

Figure 2 further visualizes the prediction quality. The predicted trajectories closely follow the ground-truth demonstrations, suggesting that the fine-tuned policy effectively captures the mapping from multimodal observations to latent actions.

Table 3: Real-world manipulation performance.

Metric	Result	Success Rate
Pickup Success	8/10	80%
Task Success	7/10	70%

4.3 Real-world Evaluation

We further deploy the fine-tuned policy on the Unitree G1 humanoid robot. During evaluation, the toy position and the robot’s initial distance are randomly perturbed within predefined ranges, while the chair position remains fixed. All experiments are conducted with an overhead gantry system to ensure operational safety.

The fine-tuned policy successfully performs the complete manipulation pipeline, including object approach, grasping, transportation, and placement. Remaining failures mainly arise from challenging initial configurations or unstable grasps, indicating that the current policy is still sensitive to distribution shifts and lacks explicit object-state feedback during execution.

4.4 Deployment Video

Representative deployment videos are available at:

https://drive.google.com/drive/folders/1saRkgkCGa_nxNbQCDxWpuXT1juYgGZnh?usp=sharing

5 Discussion

The experimental results demonstrate that task-specific fine-tuning enables a general-purpose Vision-Language-Action foundation model to perform a complete long-horizon manipulation task on a real humanoid robot. At the same time, the observed failure cases reveal several practical challenges that remain unsolved for reliable real-world deployment. In this section, we discuss the effectiveness of the proposed system, analyze representative failure modes, and summarize its current limitations.

5.1 Fine-tuning Effectiveness

The proposed fine-tuning pipeline successfully transfers the pretrained GR00T N1.7 foundation model to the target toy pickup-and-placement task. Unlike offline evaluation, which only measures prediction accuracy on demonstration data, the real-world experiments verify that the learned policy can execute the complete manipulation sequence on the physical Unitree G1 robot.

The policy achieves an 80% pickup success rate and a 70% end-to-end task success rate under randomized initial object positions. These results indicate that the collected demonstrations provide sufficient supervision for learning the overall manipulation procedure, including object approach, grasping, transportation, and placement. Moreover, successful executions under moderate scene variations suggest that the fine-tuned policy preserves a certain degree of robustness inherited from the pretrained foundation model.

Nevertheless, the current evaluation does not include a direct comparison with the original pretrained GR00T N1.7 model. Consequently, although the experiments demonstrate the feasibility of task-specific adaptation, they do not quantitatively isolate the performance improvement introduced by fine-tuning. A controlled comparison between pretrained and fine-tuned policies would provide a more comprehensive understanding of the contribution of downstream adaptation.

5.2 Failure Analysis

Although the proposed system successfully completes most evaluation trials, several representative failure modes are consistently observed during real-world deployment. Figure 3 illustrates two typical examples.

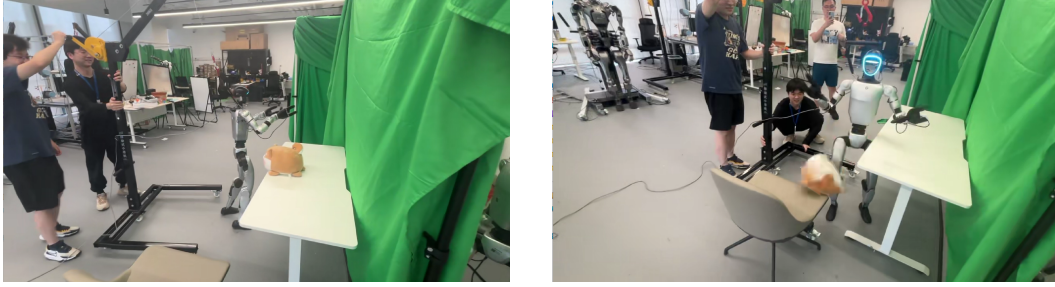


Figure 3: Representative failure cases during real-world deployment. **Left:** Small deviations in the initial robot orientation cause the target object to move outside the expected visual distribution, leading to incorrect action prediction. **Right:** The robot continues executing the learned manipulation trajectory after losing the object because the policy lacks explicit grasp-state awareness.

5.2.1 Sensitivity to Initial Conditions

One major source of failure originates from the sensitivity of the learned policy to the initial robot configuration. Before the manipulation begins, small deviations in the robot’s body orientation may cause the onboard camera to observe the scene from viewpoints that differ from those encountered during demonstration collection. As a result, the visual input gradually deviates from the training distribution, increasing the likelihood of incorrect action prediction. Since the manipulation task consists of multiple sequential stages, these early perception errors frequently propagate throughout the remaining execution and ultimately lead to task failure.

5.2.2 Lack of Explicit Grasp-state Feedback

Another common failure arises from the absence of explicit object-state reasoning. The current policy is trained through supervised behavior cloning and primarily learns to reproduce demonstrated motion trajectories. Consequently, the robot cannot explicitly determine whether the object has been successfully grasped or whether it has slipped during transportation. In several unsuccessful trials, the robot continues executing the remaining manipulation sequence even after losing the object, eventually resulting in placement failure.

5.2.3 Long-horizon Error Accumulation

Long-horizon manipulation naturally amplifies errors introduced during earlier stages of execution. For example, a slightly inaccurate grasp may still allow the robot to lift the toy, but the unstable grasp often causes the object to slip during transportation or prevents accurate placement onto the chair. Such error accumulation explains why the overall task success rate remains lower than the pickup success rate, despite the relatively reliable grasping performance.

5.3 Limitations

Despite encouraging real-world performance, the current system still exhibits several limitations.

First, the demonstration dataset is relatively small. Although the collected 128 expert demonstrations are sufficient for task-specific adaptation, they cover only a limited range of object positions, robot poses, and environmental variations. Increasing both the scale and diversity of demonstrations is expected to improve policy generalization.

Second, the current policy operates purely through supervised imitation learning without explicit state estimation or task monitoring. Unlike closed-loop planning systems, the policy cannot verify whether intermediate subtasks, such as successful grasping or object placement, have been completed before proceeding to subsequent actions.

Third, the evaluation considers only a single manipulation task involving one object category and one destination. Additional experiments involving diverse objects, cluttered environments, and more complex task compositions are necessary to evaluate the generalization capability of the learned policy.

Finally, the proposed system relies on the pretrained GR00T foundation model and the SONIC whole-body controller. Although this modular design significantly simplifies deployment, it also limits the ability to jointly optimize high-level decision making and low-level motion generation. Exploring end-to-end optimization strategies or integrating explicit feedback mechanisms into the control loop may further improve robustness for long-horizon humanoid manipulation.

6 Conclusion

In this project, we present a complete pipeline for long-horizon language-conditioned humanoid manipulation based on the GR00T N1.7 Vision-Language-Action foundation model and the SONIC whole-body controller. By collecting 128 high-quality whole-body teleoperation demonstrations and performing task-specific supervised fine-tuning, we successfully adapt a general-purpose humanoid foundation model to a real-world toy pickup-and-placement task on the Unitree G1 robot.

Experimental results demonstrate that the proposed system achieves an 80% pickup success rate and a 70% end-to-end task success rate under moderate variations in the initial object configuration. These results verify the feasibility of adapting pretrained VLA foundation models to downstream humanoid manipulation tasks through demonstration fine-tuning. Moreover, our experiments provide practical insights into deploying large robot foundation models on physical humanoid platforms.

At the same time, the observed failure cases highlight several open challenges for long-horizon robot manipulation, including sensitivity to distribution shifts, the absence of explicit object-state feedback, and accumulated execution errors over long manipulation sequences. Future work will investigate larger and more diverse demonstration datasets, explicit perception and task-state estimation, and closed-loop recovery mechanisms to further improve robustness and generalization in real-world humanoid manipulation.

Author Contributions

Yifei Zhang led the deployment of the SONIC whole-body controller on the Unitree G1 humanoid robot and was responsible for system integration, model training, and real-world evaluation. Pengfang Qian contributed to demonstration collection, model training, experimental evaluation, and project presentation. Yihai Tian participated in demonstration collection, model training, experimental evaluation, and manuscript preparation. Haocheng Xu contributed to data collection, experimental analysis, and report writing. Li Ji participated in demonstration collection, model training, real-world evaluation, and manuscript preparation. Jinhao Wu was responsible for camera server debugging, data collection, and assistance with real-world deployment.

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